Actividad metodo de la secante.

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- $f(x) = Ln(x) - 2x + 3 = 0$

$x = 1$

$f(1) = Ln(1) - 2(1) + 3 = 1 \rightarrow f(1) > 0$

$x = 2$

$f(2) = Ln(2) - 2(2) + 3 = -0.3068 \rightarrow f(2) < 0$

$X_0 = 1$; $X_1 = 2$.

$f(1) > 0$ y $f(2) < 0$

↓ continua.

$$X_2 = 2 - \frac{(-0.3069)(2-1)}{-0.3068 - 1} = 1.7651$$

$f(x_2) = Ln(1.7651) - 2(1.7651) + 3 = 0.037869$

$$X_3 = 1.7651 - \frac{(0.037868)(1.7651 - 1)}{(0.03786) - (1)} = 1.790991$$

$f(x_3) = Ln(1.7953) - 2(1.7953) + 3 = 0.000798$

$$X_4 = 1.7953 - \frac{(-0.005427)(1.7953 - 1.7651)}{(-0.005427) - (0.038)} = 1.791539$$

$f(x_4) = Ln(1.7915) - 2(1.7915) + 3 = -0.000002$

$$\varepsilon_a = \left| \frac{1.791539 - 1.790991}{1.791539} \right| \times 100 = 0.0306\% \checkmark$$

$x \approx 1.791539$.

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- $f(x) = x^3 - 3x - 4$, $x_0 = 5$ y $x_1 = 7$

$f(x_0) = 5^3 - 3(5) - 4 = 106$

$f(x_1) = 7^3 - 3\times7 - 4 = 318$

- $x_2 = 7 - \frac{318 \times (7-5)}{318 - 106} = 4 \rightarrow f(x_2) = 4^3 - 3(4) - 4 = 48$

- $x_3 = 4 - \frac{48 \times (4-7)}{48 - 318} = 3.46667 \rightarrow f(x_3) = (3.46667)^3 - 3(3.46667) - 4$

$f(x_3) = 27.26131$

- $x_4 = 3.46667 - \frac{27.2617 \times (3.4667 - 4)}{27.2617 - 48} = \boxed{2.7657}$

↳ $f(x_4) = (2.7657)^3 - 3(2.7657) - 4 = 8.859$

- $x_5 = 2.7657 - \frac{8.859 \times (2.7657 - 3.4667)}{8.859 - 27.2617} = \boxed{2.4283}$

↳ $f(x_5) = (2.4283)^3 - 3(2.4283) - 4 = 3.034$

- $x_6 = 2.4283 - \frac{3.034 \times (2.4283 - 2.7657)}{3.034 - 8.859} = \boxed{2.2525}$

↳ $f(x_6) = x_6^3 - 3(x_6) - 4 = 0.67114.$

- $x_7 = 2.2525 - \frac{0.67114 \times (2.2525 - 2.4283)}{0.67114 - 3.034} = \boxed{2.20257}$

↳ $f(x_7) = (x_7)^3 - 3(x_7) - 4 = 0.0776.$

- $x_8 = 2.20257 - \frac{0.0776 \times (2.20257 - 2.2525)}{0.0776 - 0.67114} = \boxed{2.1960}$

↳ $f(x_8) = (x_8)^3 - 3(x_8) - 4 = 0.0025$

- $x_9 = 2.1960 - \frac{0.0025 \times (2.1960 - 2.20257)}{0.0025 - 0.0776} = \boxed{2.1959}$ ✓

Ea

$Ea = \left(\frac{x_9 - x_8}{x_9}\right) \times 100 = 0.0047\%$

- $f(x) = x^2 - x + 6$, $x_0 = 1$ y $x_1 = 2$.

$f(x_0) = 1^2 - 1 + 6 = 6$; $f(x_1) = 2^2 - 2 + 6 = 8$.

- $x_2 = 2 - \frac{8 \times (2-1)}{8-6} = \boxed{-2} \rightarrow f(x_2) = -2^2 - 2 + 6 = \boxed{12}$.

- $x_3 = -2 - \frac{12 \times (-2-2)}{12-8} = \boxed{10} \rightarrow f(x_3) = x_3^2 - x_3 + 6 = \boxed{96}$

- $x_4 = 10 - \frac{96 \times (10 - -2)}{96 - 12} = \boxed{-3.7143} \rightarrow f(x_4) = x_4^2 - x_4 + 6 = \boxed{23.5102}$

- $x_5 = -3.7143 - \frac{23.5102 (-3.7143 - 10)}{23.5102 - 96} = \boxed{-8.1622} \rightarrow f(x_5) = \boxed{80.793}$

- $x_6 = -8.1622 - \frac{80.793 \times (-8.1622 - -3.7143)}{80.793 - 23.5102} = \boxed{-1.9994}$.

↳ $f(x_6) = (x_6)^2 - x_6 + 6 = \boxed{11.4547}$.

- $x_7 = -1.9994 - \frac{11.4547 \times (-1.9994 - -8.1622)}{11.4547 - 80.793} = \boxed{-0.8519}$.

↳ $f(x_7) = (x_7)^2 - x_7 + 6 = \boxed{7.5776}$.

- $x_8 = -0.8519 - \frac{7.5776 \times (-0.8519 - -1.9994)}{7.5776 - 11.4547} = \boxed{1.174023}$

R+A / con el metodo del secante no encuentra una solución real ya la ecuación no tiene raices reales

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- $f(x) = 2\operatorname{sen} x - x$; $x_0 = 1$ y $x_1 = 2$.

$f(x_0) = 2\operatorname{sen}(1) - 1 = 0.682942$

$f(x_1) = 2\operatorname{sen}(2) - 2 = -0.181405$

$$x_2 = 2 - \frac{-0.181405 \times (2-1)}{-0.181405 - 0.682942} = \boxed{1.790125}$$

$$f(x_2) = 2\operatorname{sen} x_2 - x_2 = \boxed{0.161963}$$

$$x_3 = 1.790125 - \frac{0.161963 \times (1.790125 - 2)}{0.161963 - -0.181405} = \boxed{1.989121}$$

↳ $f(x_3) = 2\operatorname{sen} x_3 - x_3 = 0.010402$.

$$x_4 = 1.989121 - \frac{0.010402 \times (1.989121 - 1.790125)}{0.010402 - 0.161963} = \boxed{1.995915}$$

↳ $f(x_4) = 2\operatorname{sen} x_4 - x_4 = -0.000699$

$$x_5 = 1.995915 - \frac{-0.000699 \times (1.995915 - 1.989121)}{-0.000699 - 0.010402} = \boxed{1.995493}$$ ✓

$$Ea = \left| \frac{1.995493 - 1.995915}{1.995493} \right| \times 100 = 0.0223\ \%$$ ✓